Activity: Computer Forensics

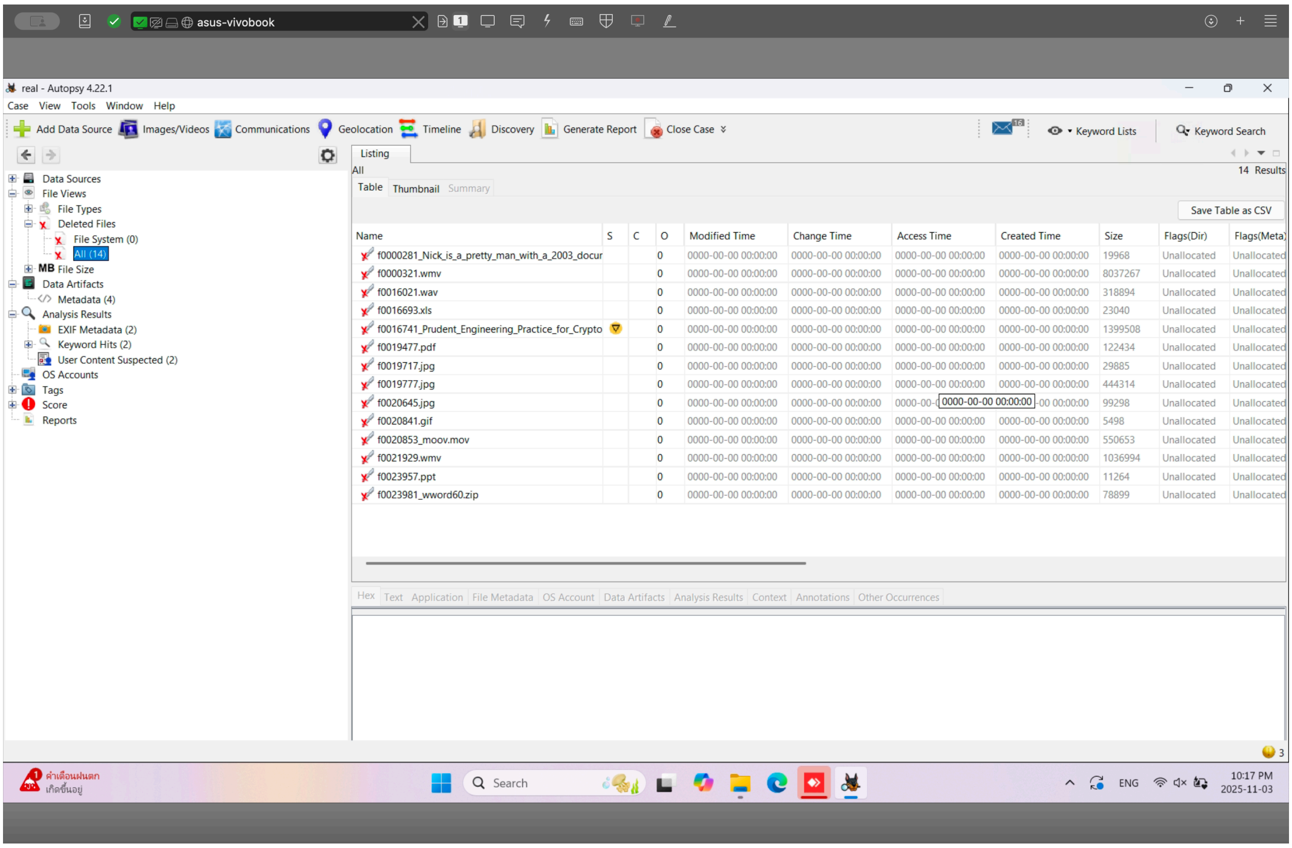
Part I. File Carving

1. Look at the data on the file system (Click on Data Sources and look at the hex values on the right). The file system has no files, but why are we able to find items on the disk image? Explain why the file system has no files but there are items that can be found on the disk image.
Ans:

When a user issues a delete command, the operating system only deletes or destroys the metadata and reports that the space is now "free". However, deleting metadata does not mean that the actual content of the file (file content) is removed from the disk sectors. The raw data of the file still exists in the "free" space until new files are written over it.

2. How many objects can you find?
Ans: 14

3. List all the objects here and report on whether or not the content is accessible or damaged/ corrupted. Also note which files were actually already deleted.
Ans: Only image and text files can be opened. Audio files cannot be opened in Autopsy, but they are accessible in external tools

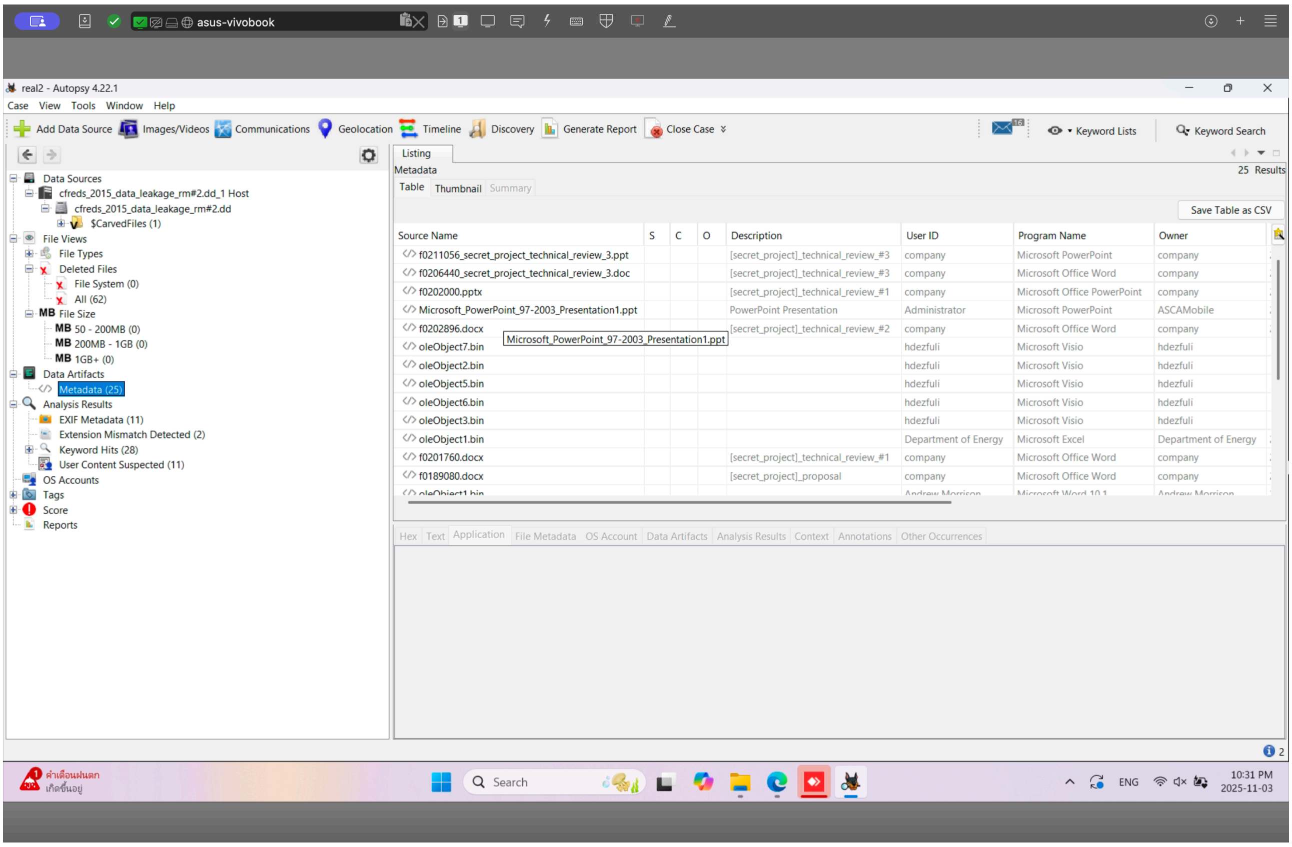


4. Think securely: If we want to delete files on a magnetic hard disk and not have them be recovered by any tool, what do we need to do? And how much time do you think you need to wipe a 1TB magnetic hard disk?
Ans: Writing over the disk at least 7 times will take approximately 14 hours, or using a strong magnet for a few seconds.

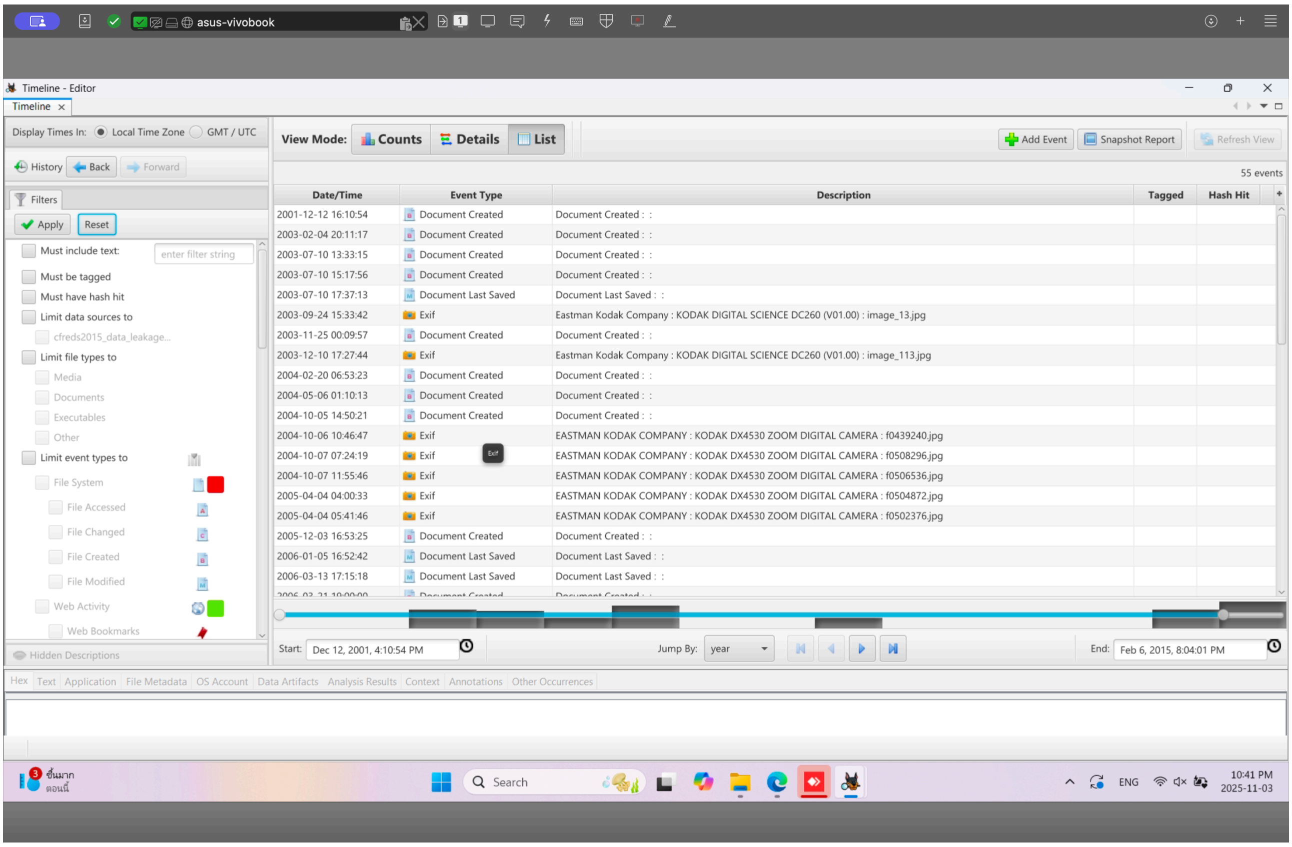
5. Will file carving be able to recover deleted files on an SSD? Why or why not?
Ans: File carving cannot recover deleted files on an SSD, or it will be very difficult. SSDs use the TRIM or UNMAP command, and the SSD controller responds by clearing the actual data.

Part II. Investigation

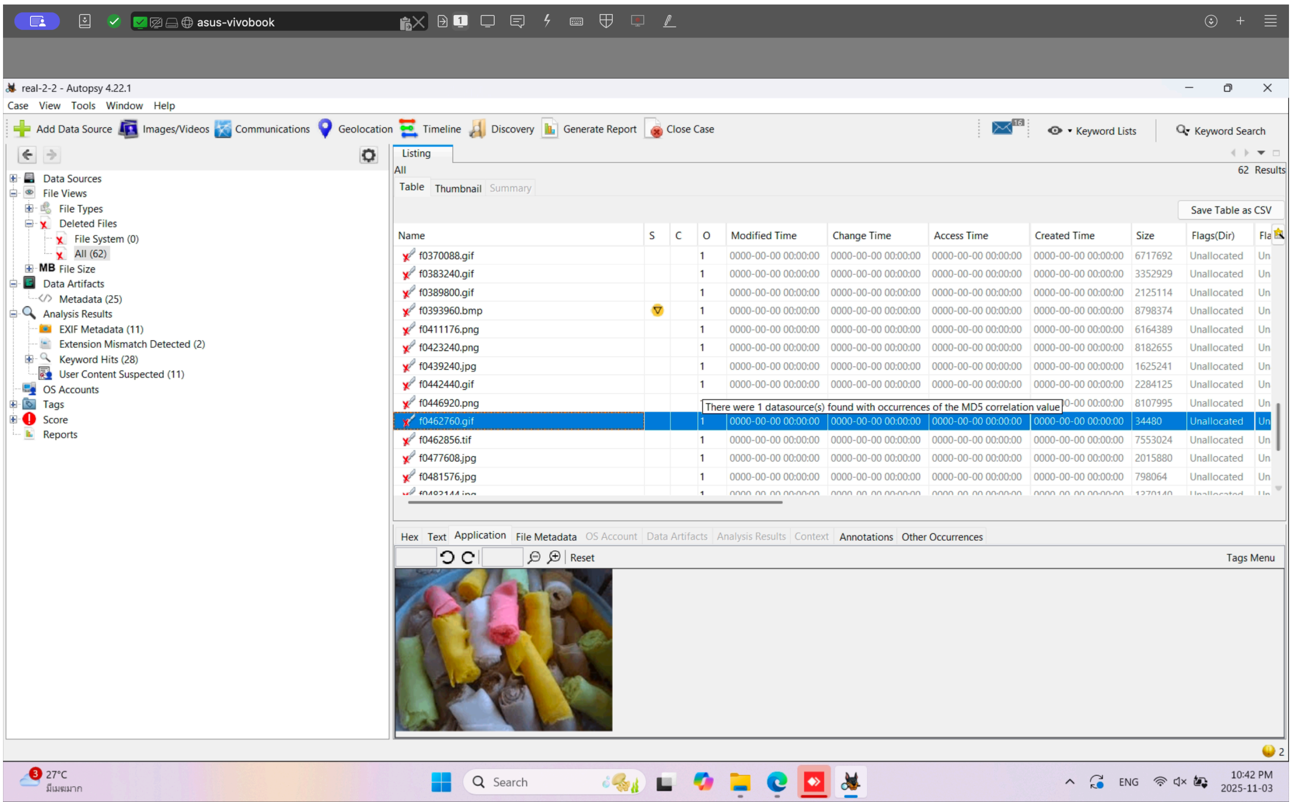
1. List all directories that were traversed in ‘RM#2’.
Ans: CarvedFiles

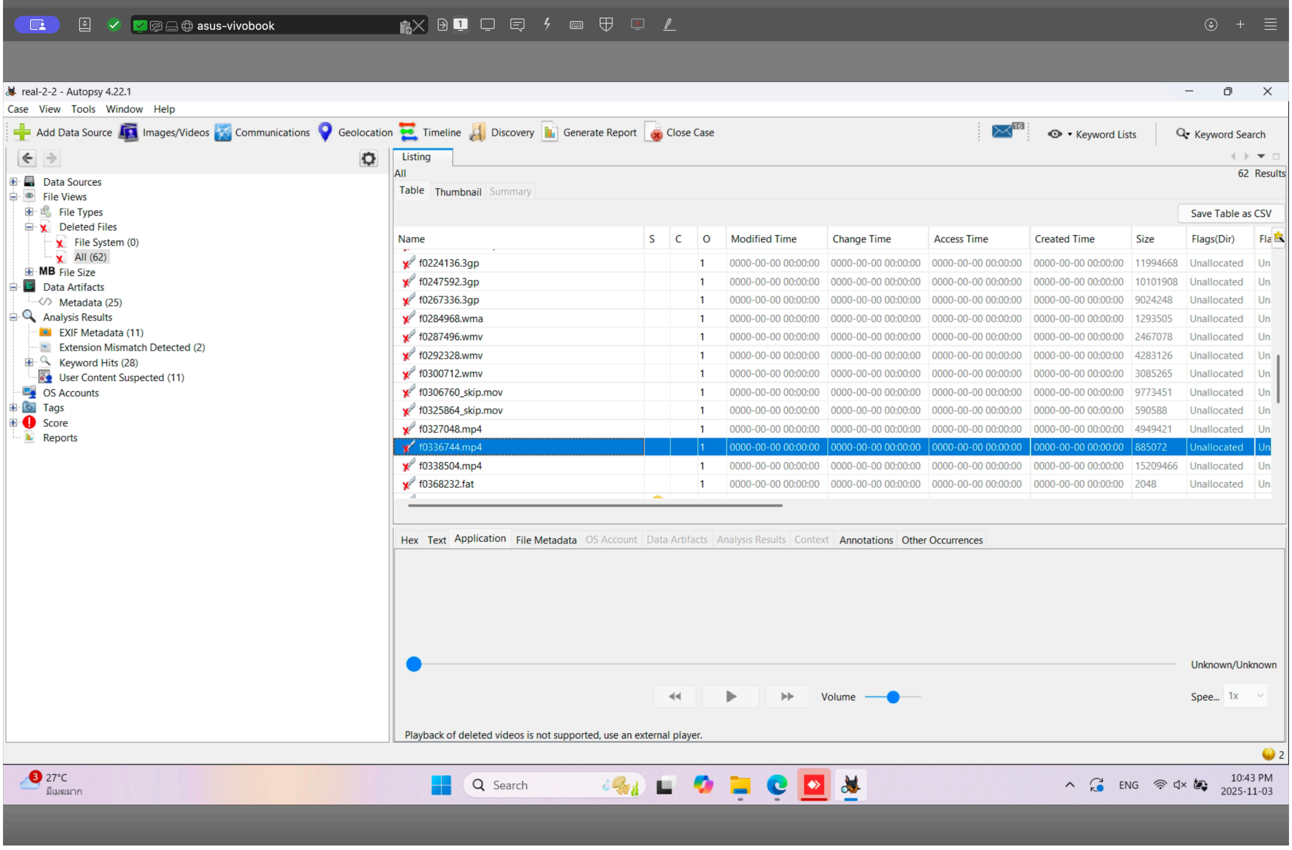


2. List all files that were opened in 'RM#2'.
Ans:



3. Recover deleted files from USB drive ‘RM#2’. What files were you able to recover?
Ans: Most images could be opened, but MP4 files could not.

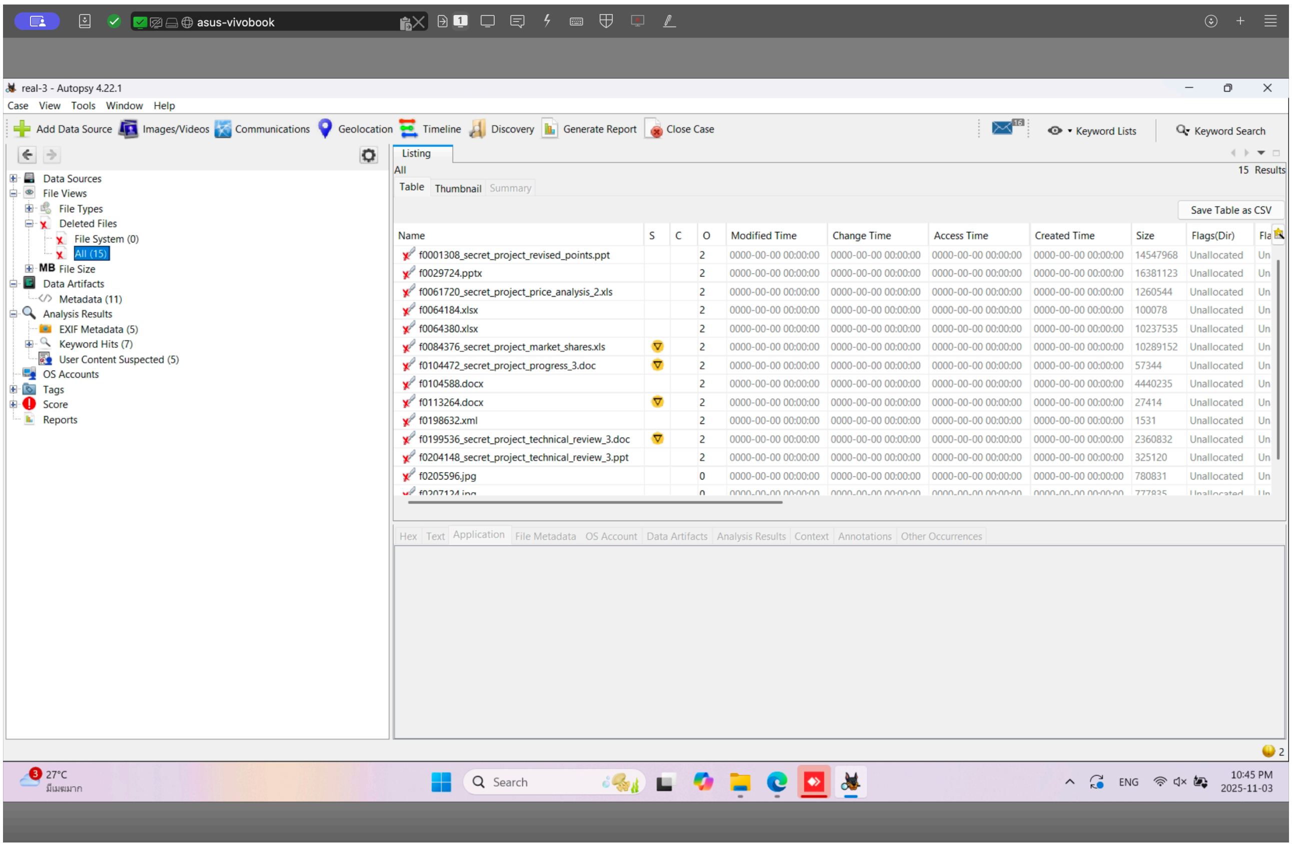




4. What actions were performed for anti-forensics on USB drive 'RM#2'? [Hint: this can be inferred from the results of the above question]
Ans:

There was an attempt to delete the metadata, and some file formats were tampered with. Notably, there were files with strange names ending in .txt, but the content inside didn't seem to be text, so I tried changing the type to .ppt.

5. Recover hidden files from the CD-R 'RM#3'. What files were you able to recover?
Ans: Found 15 files.



6. What actions were performed for anti-forensics (data hiding) on CD-R 'RM#3'?
Ans: It seems that only the metadata was deleted or destroyed.